

Lingam Hemang

+91 6302922329 hemanglingam@gmail.com [LinkedIn](#) [GitHub](#) [Portfolio](#)

Professional Summary

Python Engineer with **4 years** at Infosys building **production data systems, backend APIs, and GenAI applications** for **Apple Inc.** – on a proprietary platform built on **Spark, Kafka, Airflow, and AWS**. Proven impact across critical data flows: **40% faster pipelines, 99.5% uptime, 35% fewer errors**. Experienced across ETL engineering, semantic modelling, REST API integration, platform tooling, and backend data support. Actively building **GenAI applications** using LangChain, RAG pipelines, and AWS Bedrock.

Education

SRM Institute of Science and Technology

May 2018 – June 2022

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 9.52/10.0

Technical Skills

Languages: Python, SQL, PySpark

Backend & APIs: Django, Flask, REST API Design & Integration, Server-Sent Events (SSE), OOP, SOLID Principles, Design Patterns, pytest

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Semantic Layer & Materialisation, Data Modelling, Schema Design & Validation, Data Quality, Partitioning & Query Optimisation, Tableau

Data Formats: JSON, Parquet, Avro, CSV

GenAI / ML: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, XGBoost, LightGBM, scikit-learn, Plotly, Dash, LLM, Prompt Engineering

Tools & Platform: Proprietary Spark/Kafka/Airflow/AWS Data Platform, Splunk, Git, Agile/Scrum, Jira, Confluence

Experience

Infosys

Hyderabad, India

Specialist Programmer – Data Engineer

July 2022 – Present

Client: **Apple Inc.** | Python, Spark, Kafka, Airflow, Snowflake, AWS S3, PySpark, Splunk, Tableau

- Engineered production **ETL pipelines** on a proprietary data platform built on **Apache Spark, Kafka, Airflow, and AWS**; consumed internal and external **REST APIs** (in-house team endpoints, Pharos) via Python to ingest data into **Snowflake**; **distributed processing and SQL query optimisation** cut end-to-end processing time by **40%**.
- Built and optimised **semantic layer processes** using Python and database connector packages; **materialised tables and views** for downstream consumption, significantly improving **Tableau dashboard** query performance; supported multiple **Tableau datasources** and dashboards for business analytics and operational reporting.
- Designed **high-availability, fault-tolerant ingestion pipelines** across **AWS S3** and **Kafka event streams** via **Airflow DAGs** with retry logic and SLA monitoring – achieving **99.5% uptime**; schema validation and self-healing scripts reduced pipeline error rate by **35%**; **deployed a client-provided ML name-matching model** to the Airflow cluster using an in-house deployment tool – provisioning input views and configuring Airflow connections for model execution.
- Developed **reusable in-house framework extensions** for new data source integrations reducing onboarding effort by **30%**; extended the **Splunk connector**, built custom queries to load usage data into Snowflake, and created a **Tableau security audit dashboard** to monitor data access patterns and identify unauthorised access gaps.
- Provided **backend data support** for a production application – maintaining **data consistency in referencing database views** and loading data into a **semantic search cluster**; applied data governance standards and collaborated with engineers and BAs across Agile feature teams via Jira and Confluence.

Projects

Agentic AI Git Diff Analyzer | Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab | [Repo](#)

- Dual-mode SQL schema impact analyzer** – local (Ollama) and **AWS mode** (Bedrock Claude 3.5 via GitHub Webhook → Lambda → SSM → EC2); commit-accurate SQL scanning with downstream dependency detection across the repo via regex; LLM impact reports with **risk-level badges** (CRITICAL → LOW) as styled A4 PDFs (ReportLab); real-time **Django UI** with **SSE** for live run status and run deduplication by commit hash.

Document Intelligence Chatbot – RAG Pipeline | Python, Django, LangChain, ChromaDB, Ollama, EasyOCR | [Repo](#)

- End-to-end **RAG pipeline** (LangChain + ChromaDB + Ollama) with persistent multi-turn conversation memory; multi-format ingestion (PDF, DOCX, TXT, Images) via **PyMuPDF + EasyOCR** with auto-rotation; incremental vector store with **timestamped snapshots/rollback**; **10-endpoint Django REST API** supporting live model switching and VectorDB management.

Retail ML Sales Forecasting API | Python, Flask, scikit-learn, Plotly, Dash | [Repo](#) **Infosys Hackathon – 2nd Place**

- Multi-scenario **ML forecasting system** (Random Forest, XGBoost, LightGBM) for sales, store projections, and product recommendations via **Flask REST API**; cyclical sin/cos month encoding for seasonal patterns; cross-join **purchase probability scoring** across full product catalog; interactive **Dash + Plotly** dashboard with live reactive callbacks.